

Treadmill desks: A 1-year prospective trial.

OBJECTIVE: Sedentariness is associated with weight gain and obesity. A treadmill desk is the combination of a standing desk and a treadmill that allow employees to work while walking at low speed.

DESIGN AND METHODS: The hypothesis was that a 1-year intervention with treadmill desks is associated with an increase in employee daily physical activity (summation of all activity per minute) and a decrease in daily sedentary time (zero activity). Employees ($n = 36$; 25 women, 11 men) with sedentary jobs (87 ± 27 kg, $BMI 29 \pm 7$ kg/m²), $n = 10$ Lean $BMI < 25$ kg/m², $n = 15$ Overweight $25 < BMI < 30$ kg/m², $n = 11$ Obese $BMI > 30$ kg/m²) volunteered to have their traditional desk replaced with a treadmill desk to promote physical activity for 1 year.

RESULTS: Daily physical activity (using accelerometers), work performance, body composition, and blood variables were measured at Baseline and 6 and 12 months after the treadmill desk intervention. Subjects who used the treadmill desk increased daily physical activity from baseline $3,353 \pm 1,802$ activity units (AU)/day to, at 6 months, $4,460 \pm 2,376$ AU/day ($P < 0.001$), and at 12 months, $4,205 \pm 2,238$ AU/day ($P < 0.001$). Access to the treadmill desks was associated with significant decreases in daily sedentary time (zero activity) from at baseline $1,020 \pm 75$ min/day to, at 6 months, 929 ± 84 min/day ($P < 0.001$), and at 12 months, 978 ± 95 min/day ($P < 0.001$). For the whole group, weight loss averaged 1.4 ± 3.3 kg ($P < 0.05$). Weight loss for obese subjects was 2.3 ± 3.5 kg ($P < 0.03$). Access to the treadmill desks was associated with increased daily physical activity compared to traditional chair-based desks; their deployment was not associated with altered performance. For the 36 participants, fat mass did not change significantly, however, those who lost weight ($n = 22$) lost 3.4 ± 5.4 kg ($P < 0.001$) of fat mass. Weight loss was greatest in people with obesity.

CONCLUSIONS: Access to treadmill desks may improve the health of office workers without affecting work performance. Copyright © 2012 The Obesity Society.